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Quantitative Methods - (6–9%, ~13 Qns)

(Kindly note the number of questions range is taken from the CFA Institute website & the approximate number of questions is just an approximate middle number of x-y% of 180)

1 - Rates & Returns

- Interest Rate - synonymous terms
- Risk-free rate (nominal = real + inflation)
- Risks impacting R.o.R (maturity, default, liquidity)
- Holding Period Return (HPR)
- Compounded HPR over time 't'
- Arithmetic Mean, Geometric Mean, Harmonic Mean
 - $AM \geq GM$
 - $AM > GM > HM$
 - $AM \times HM = GM^2$ (for 2 observations)
- Geometric mean vs Annualized returns
- Money-weighted R.o.R
- Time-weighted R.o.R
- Annualized return - HPR formula
- Impact of increasing compounding frequency
- Continuous compounding
- Real Return
- Leveraged Return

2 - Time Value of Money

- Formula - PV, PV (continuous compounding)
- Price investor pays (YTM, Maturity)
- Negative IR, Negative YTM zero-coupon bond
- Perpetual bond
- Annuity bond
- Preferred stock (fixed dividend till infinity)
- Common stock
- Gordon Growth Model (GGM)
- Multi-stage GGM
- CF Additivity Principle
- No-arbitrage principle

- Forward currency exchange rate
- Relationship of Forward & Spot
- One-period Binomial model (finding C_0 using R_u , R_d , C_u , C_d , h , V_u , V_d , C_0)

3 - Statistical Measure of Asset Returns

- Central tendency, Mean, Median (*not affected by outliers vs mean*), Mode
- Modal interval
- Trimmed Mean
- Winsorized mean
- Dispersion, Relative Dispersion
- Mean Absolute Deviation
- Sample variance
- Coefficient of variation (*measures relative dispersion*)
- Target downside deviation
- Skewness - *positively (right)*, *negatively (left)*
- Kurtosis - leptokurtic ($k > 3$, + excess kurtosis), platykurtic ($k < 3$, - excess kurtosis), mesokurtic ($k = 3$)
- Covariance
- Correlation
- Spurious correlation

4 - Probability Trees & Conditional Expectations

- Expected Value
- Volatility
- Variance - mean & expected value formula
- Standard Deviation
- Bayes Theorem

5 - Portfolio Mathematics

- Expected Return
- Covariance
- Covariance Matrix
- Portfolio Variance for 2 & 3 assets - (in terms of variance, standard deviation, correlation)
- Correlation
- Shortfall Risk
- Roy's Safety First criteria
- Safety First Ratio

6 - Simulation Methods

- Lognormal Distribution
- Returns distribution - independent, identical
- Monte Carlo Simulation
 - *Identify risk variables → assign probabilities, parameters → random generation of risk factor using pricing model to value a security → Repeat 1000s times → Distribution of simulated values tells about mean & variance*
 - *Answers “what-if” scenarios*
 - *It is a statistical simulation & NOT an analytical one*
- Bootstrap Resampling
 - *Repeatedly drawing from the same sample → observation can appear multiple times*
 - *Uses actual observed data (constrained by historical distribution)*
 - ***Shows statistical features of a population from a single sample***

7 - Estimates & Inferences

- Simple Random Sampling
- Systematic Sampling
- Stratified Random Sampling
- Cluster Sampling
- Convenience Sampling
- Judgmental Sampling
- Central Limit Theorem
- Standard Error
- Resampling - Jackknife, Bootstrap
- Confidence interval (point estimate, reliability factor, standard error)
- Sampling Error
- Sampling Distribution

8 - Hypothesis Testing

- Null Hypothesis
- T-statistic
- Type 1 Error (*significance level, = 1 - confidence level*)
- Type 2 Error, Power of Test
- Level of significance
- P-value
- One-tailed test (*stronger belief of the researcher than a two-tailed, critical value lesser than two-tailed*)
- Two-tailed

- Null Hypothesis for Two-tailed & One-tailed
- Difference of mean test, paired comparison test
- Chi-square test
- F-Test (F-statistic)
- Parametric Tests
- Non-parametric Tests

9 - Parametric, Non-Parametric Test of Independence

- Correlation
- Pearson Test (*to test if population correlation = 0*)
- Spearman Rank Correlation Test (*to test if two ranks are correlated*)
- Contingent Table Data

10 - Simple Linear Regression

- SLR Equation
- Linear Regression Assumptions
- ANOVA (*learn how to analyze the ANOVA table*)
- SST ($SST = SSR + SSE$)
- SSR (*explains variation in DV basis IV*)
- MSR ($MSR = SSR$)
- SSE (*unexplained variation in DV*)
- MSE ($MSE = SSE/n-2$)
- SEE (*Standard Error of Estimates = sqrt MSE*)
- Coefficient of Determination ($R^2 = SSR/SST$)
- Correlation (R)

11 - Introduction to Big Data Techniques

- Fintech
- Big data - data sources (+corporate exhaust, IoT)
- Big data - characteristics (volume, velocity, variety)
- Data processing steps
- Neural Network
- Machine learning (training, validation, testing)
- Supervised learning
- Unsupervised learning
- Overfitting
- Underfitting
- Black box output (*outcomes with unexplainable relationships*)

- Text Analytics, NLP, Algo Trading
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Economics - (6–9%, ~13 Qns)

12 - The Firm & The Market Structures

- Short Run & Long Run - (w.r.t. factors of production)
- Perfect Competition - Shutdown & Breakeven
- Imperfect Competition - Shutdown & Breakeven
- ATC Curve & Economies - Diseconomies of Scale
- Relationship between MC & AVC curves
- Characteristics of 4 Market Structures (Perfect competition, Monopolistic competition, Oligopoly, Monopoly)
- Monopolistic Competition - Output in the Short Run
- Monopolistic Competition - Output in the Long Run ($P = ATC$)
- Perfect Competition ($P=MC=MR=ATC$)
- Kinked Demand Curve model
- Cournot model
- Stackelberg model
- Dominant Firm model
- Nash Equilibrium
- N-firm concentration ratio
- Herfindahl-Hirschman Index

13 - Understand Business Cycles

- 4 components of a Business Cycle (Expansion, Peak, Contraction, Trough)
- Growth cycle & Growth rate cycle
- Credit cycle
- Inventory-Sales ratio in Expansion & Contraction
- Consumer spending patterns (Durable Goods, Services, Non-Durable Goods, Housing)
- Domestic GDP - Import/Export relationship
- Currency - Import/Export relationship
- Economic Indicators for Peaks & Troughs (Leading, Coincident, Lagging)

14 - Fiscal Policy

- Fiscal Policy & its objective
- Fiscal surplus
- Fiscal deficit & benefits
- Discretionary fiscal policy
- Country's debt ratio
- Cons of fiscal deficit
- Crowding-out effect

- Fiscal Policy tools (Expenditure - Transfer of Payment, Current, Capital & Revenue - Direct & Indirect Taxes)
- Desirable attributes of an FP (SEFS)
- Marginal Propensity to Consume
- Fiscal Multiplier
- Balanced Budget Multiplier
- Ricardian Equivalence
- Lags in FP changes
- Structural deficit / Cyclically Adjusted budget deficit

15 - Monetary Policy

- Monetary Policy & its objective
- Key roles of the Central Bank
- Menu costs & Shoe-Leather costs due to High Inflation
- Target inflation rate
- Pegging
- Monetary Policy tools (Open Market Operations, Reserve Requirements, Policy Rate)
- Monetary Transmission Mechanism
- Impact of Policy Rate (Monetary Policy, ST Lending rates, Aggregate demand, Credit purchases, asset prices, consumer expenditures, foreign investment in debt securities, currencies)
- 3 Qualities of the Central bank
- Exchange Rate targeting, net effect of ERT
- Neutral interest rate
- Quantitative easing
- Impact of MP & FP - Interest Rates, Output, Private spending & Public spending

16 - Geopolitics

- Geopolitics, Globalization, Nationalism
- Archetypes of Globalization & Co-operation (Bilateralism, Multilateralism, Autarky, Hegemony)
- WTO
- World Bank
- IMF
- 3 Geopolitical Risks (Event, Exogenous, Thematic)
- Black Swan Risk
- Velocity Risks
- 3 Geopolitical Tools (National Security, Economic, Financial)

17 - International Trade

- Trading Blocs or Regional Trade Agreements (FTA, Customs Union, Common Markets, Economic Union, Monetary Union)
- Trade Restrictions (Tariffs, Quotas, Export Subsidies, Voluntary Export Restraints)
- Effect of Tariffs & Quotas
- Deadweight loss, Quota rents

18 - Capital Flows & FX Market

- Hedging, Speculation
- Sell & Buy side FX players
- Price currency & Base Currency
- Sums - % change in FX value of currency
- Real P/B Exchange Rate
- Spot Exchange Rate, Forward Rate
- ER Regimes for countries WITHOUT their own Currency -
 - Formal Dollarization
 - Monetary Union
- ER Regimes for countries WITH their own Currency -
 - Currency Board Arrangement
 - Pegging - Conventional fixed agreement
 - Pegging - Horizontal bands
 - Pegging - Crawling
 - Crawling Bands
 - Managed Floating
 - Independent Floating
- Capital Flows
- Objectives of Capital Flows
- ST & LT ER determinants

19 - Exchange Rate Calculations

- Cross Rate
 - Sums - Cross Rate
 - Sums - No-Arbitrage Forward Exchange Rate
 - Sums - No-Arbitrage Forward Exchange Rate w 90/180-day IR
 - Arbitrage Profit
 - Sums - Forward Exchange Rate with Points
 - Sums - Forward Exchange Rate with Percentage
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Corporate Issuers - (6–9%, ~13 Qns)

20 - Organizational Forms, Corporate Issuer Features, Ownership

- 4 Types of Business Structures (Separate Legal Entity?, Managed by?, Extent of Owner Liability, Access to Capital, Taxation on Profits)
 - *(Sole Proprietorship, General P., Limited P., Corporations)* -
- Limited Liability Partnerships (LLP)
- Double Taxation
- Article of Incorporation
- Free Float
- Private Placement
- 3 ways to go public - IPO, Direct listing, SPACs
- Why co. goes from public to private? (done via LBOs)
- Sums - Effective Tax Rate

21 - Investors & Other Stakeholders

- Debt holders vs Equity holders (*equity holders/shareholders will want the co. to take on risk due to unlimited upside potential, while debt holders/creditors won't*)
- Shareholder theory (*Maximize market value of common equity*)
- 2 Types of Lenders - Bond holders & Private debt holders
- Board structure - One-tier & Two-tier
- Staggered board
- Stranded assets
- Leverage & impact on ROE
- Sum - ROE

22 - Corporate Governance: Conflicts, Mechanisms, Risks & Benefits

- Principal-Agent Conflict
- Dual-class structure of votes
- Proxy voting
- Bond indenture
- Activist shareholder (Proxy contests & Tender offer)
- Protect against Hostile takeovers (Staggered board, Poison Pill provision)
- Stakeholder theory (resolving conflicts of interest between owners & creditors)

23 - Working Capital & Liquidity

- WC Cycle (*Cash* → *Purchase RM* → *ACCOUNTS PAYABLE* → *Manufacturing* → *INVENTORY* → *Sales* → *ACCOUNTS RECEIVABLES* → *Collections* → *Cash*)
- Cash Conversion Cycle - Formula, Impact on Liquidity
- 2 types of Liquidity - Drag (inflow lag) & Pull (outflow accelerated)
- Formula - Total, Net Working Capital
- 2 sources of Liquidity - Primary & Secondary
- Liquidity Ratios - Current, Quick (only Cash, MS & AR), Cash (only Cash & MS)
- WCM approaches -
 - **Conservative** (High current assets → more liquidity, financing WC using LT debt & equity → more permanent capital → higher costs → less profits, less financial risk)
 - **Aggressive** (Low CA → lesser liquidity, financing WC using ST debt → lower costs → higher returns, vulnerable to business/market disruptions)
- Sums - Annual cost of trade credit (EAR), Liquidity Ratios

24 - Capital Investment & Allocation

- 4 types of capital investments/projects - Going concern, Regulatory/Compliance, Expansion, Other.
- 4 steps of capital allocation - idea generation, analyzing firm-wide projects (CF forecast), budget creation, monitoring decisions & post audits.
- **NPV** - formula, (+ve → *accept independent projects, direct measure of expected increase in firm's value*)
- **IRR** - formula, (IRR>WACC → accept project), disadvantages
- Hurdle rate
- Conventional & unconventional CF patterns (Multiple/No IRR)
- *IRR better than NPV* → MoS
- **Return on Invested Capital (RoIC)** - formula, pros.
- Sums - NPV, IRR
- Principles of Capital Allocation
- Cannibalization
- Positive Externality
- Two types of Errors while evaluating projects - Cognitive & Behavioural
- Real options - Timing, Abandonment, Expansion, Flexibility (price & production), Fundamentals

25 - Capital Structure

- WACC
- Factors affecting the proportion of debt in the capital structure
- Ratios to assess debt capacity (Interest coverage, Leverage)
- Capital structure for different company stages (startup, growth, mature)

- Sums - WACC
- MM 1, MM 2 Propositions
- Static Tradeoff theory
- Free CF hypothesis
- Pecking Order theory

26 - Business Models

- Business Model components
 - Pricing Strategies
 - Pricing Models
 - Value proposition
 - Value chain
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FSA - (11–14%, ~22 Qns)

27 - Introduction to Financial Statement Analysis

- FSA Framework
- Purpose of Financial Reporting & FSA
- Standard Setting Boards - FASB, IASB
- Regulatory Authorities - SEC, FCA, IOSCO
- Proxy Statements
- Financial Statement Notes / Footnotes
- SEC Filings - 10K, 10Q, 8K, 14A, 144, 3-4-5, S1
- MDA
- Data sources for FS - insider, public TP, proprietary TP, primary research
- Audit & opinions - unqualified, qualified, adverse, disclaimer

28 - Analyzing Income Statement

- **Revenue Recognition**
 - Performance Obligation & Progress towards completion
 - Agenting
 - Service vs Licensing
 - Franchising
- **Expense Recognition**
 - Matching Principle
 - Period costs
 - Capitalization vs Expensing
- **Capitalization vs Expensing** effects on IS line items
- Capitalized interest - reporting on IS, CF, Sums
- Non-Recurring items
 - Reporting of Unusual/infrequent items
 - Reporting of Discontinued
 - Retrospective vs prospective application
- **Ratios & Common Sizing of IS**
 - IS Ratios - GPM, NPM, OPM
- **Earnings Per Share**
 - Basic EPS
 - Diluted EPS
 - Sums - WA shares outstanding
 - Sums - EPS with convertible preferred stock
 - Sums - EPS with convertible bond
 - Sums - EPS with warrants/ESOPs

29 - Analyzing Balance Sheet

- **Intangible Assets**
 - What are intangible assets?
 - Intangible asset classification - identifiable, unidentifiable
 - Reporting of intangible assets if purchased & if created internally.
 - Reporting of finite & indefinite lived intangible assets
 - Items "Expensed as Incurred" under both GAAP & IFRS
 - Research & Development → Expensing & Capitalizing
- **Goodwill**
 - What is goodwill?
 - Goodwill formula
 - Reporting of goodwill
 - Effect of goodwill impairment on co. value, IS, CF
- **Financial Reporting**
 - IFRS/GAAP - AC/HTM, FVOCI/AFS, FVPL/TS
 - Ammortized Cost - (*debt securities HTM, loans payable, notes payable, unlisted equity | unrealized P&L not considered*)
 - FVOCI - (*debt securities before maturity, equity (if classified) | unrealized P&L on OCI*)
 - FVPL - (*debt securities ST, equity, derivatives | unrealized P&L on IS*)
- **Financial Reporting of Non-Current Liabilities**
- BS Common Sizing
- Liquidity Ratios - current, quick, cash
- Solvency Ratios - debt, financial leverage

30 - Analyzing Statement of Cash Flows 1

- Quality of Earnings - CFO vs NI
- Asset/Liability - CFO/CFI/CFF relationship
- Sources & Uses of cash
- Accounts Receivable formula
- CFO - Direct
- CFO - Indirect
- CFI Activities
- CFF Activities
- Indirect method to direct method

31 - Analyzing Statement of Cash Flows 2

- Free Cash Flow to the Firm
- Free Cash Flow to the Equity

- Performance Ratios - CF to Revenue, CF to ROA, CF to ROE, Cash to Income (use operating income), CF per share
- Coverage Ratios - ICR, DCR, Reinvestment (use cash paid for long-term assets), Debt Payment (cash paid for LT debt), Dividend Payment, Investing & Financing Ratio

32 - Analysis of Inventories

- Inventory Measurement
 - Net Realizable Value
 - IFRS - $\text{MIN}(\text{cost}, \text{NRV})$
 - GAAP - $\text{MIN}(\text{cost}, \text{market}), \text{market} \rightarrow \text{NRV}, \text{NRV-NPM}$
- Impact of Inventory write-down (as cost of sales) - in write-down periods & subsequent periods
- Impact of Inflation/Deflation on Inventory Costing Methods (LIFO, FIFO)
- LIFO, FIFO w.r.t COGS & closing inventory
 - *LIFO* → Lesser write-downs (due to pricing based on oldest inventory)
 - *FIFO* → best current estimate of closing inventory (due to inventory that comes in recently does not go out first and thus contributes more to closing inventory)
- LIFO Liquidation
- Presentation & Disclosure
 - Required inventory disclosures
 - Raw material, WIP, Finished Goods, Sales relationship
 - Firm's quality of inventory management (Inventory Turnover & Sales relationship)

33 - Analysis of Long-Term Assets

- **Intangible & Long Lived Assets**
 - Reporting of impairment of indefinite-lived assets on IS
 - Intangible assets - identifiable & unidentifiable
 - Reporting of Intangible Assets - Created Internally
 - Reporting of Intangible Assets - Purchased
- **Impairment**
 - Impairment under IFRS - *carrying value vs recoverable amount (FV-SC, value)*
 - Impairment under GAAP - *carrying value vs undiscounted future CFs*
 - Impact of impairment on FS line items (*loss on IS, asset low on BS, no impact on CF*)
 - Held-for-sale asset
- Derecognition
 - Reporting of Long-Lived Assets - sold, abandoned, exchanged
- Spin-off
- Sums - Useful Life (Average Age, Total Useful Life, Remaining Useful Life)

34 - Long-Term Liabilities & Equity

- **Leases**
 - Advantages of leases over purchasing an asset
 - Financial lease vs operating lease
 - **Lessee Accounting - Financial Lease & Operating Lease**
 - **ROU vs Lease Liability**
 - **Impact on the CF Statement, IS**
 - Lease Accounting for - ST & Low Value
 - **Lessor Accounting - Financial Lease & Operating Lease**
- **Deferred Compensation & Disclosures**
 - Pension Plans -
 - Defined Contribution Plan
 - Defined Benefit Plan (Plan Asset, Plan Liability/Pension Obligation)
 - Funded Status
 - IFRS Accounting for DBP (service costs, net interest income/expense, remeasurements)
 - GAAP Accounting for DBP (current service costs, interest income/expense, expected ROA, past service costs, actuarial gains/losses)
- **Shares-based Compensation**
 - Stock options
 - Stock Appreciation Rights

35 - Analysis of Income Taxes

- Core idea of DTL & DTA
- DTA (Tax expense < Tax Payable, When is a DTA created?)
- DTL (Tax expense > Tax Payable, When is a DTL created?)
- Carrying Value vs Tax Base → DTL, DTA → for Asset & Liability
- Permanent Timing Difference
- Tax Expense
- Impact of changes in Tax rate on DTA, DTL
- Realizability of DTA (IFRS; GAAP - valuation allowance)
- Analytical treatment of DTL (reversible, irreversible)
- Tax Rates (statutory tax rate, effective tax rate, cash tax rate)
- Temporary Timing Difference & creation of DTL/DTA

36 - Financial Reporting Quality

- High Quality Financial Reporting

- Quality of Reported Earnings
- Aggressive vs Conservative approach of FR
- Earning Smoothing
- Conservative Bias in FR
- Low Quality FR
- Accounting choices used to manage Earnings (revenue recognition timing, uncollectable credit, valuation allowance, depreciation method, amortisation-impairment, inventory costing method, capitalization, related party transaction)
- GAAP - IFRA Reporting of interest, dividend & tax

37 - Financial Analysis Techniques

- Common size analysis - vertical, horizontal
- Activity Ratios
- Liquidity Ratios
- Solvency Ratios
- Profitability Ratios
- Dupont Ratio - 3 & 5-way
- Industry-specific ratios

38 - Introduction to Financial Statement Modelling

- Sales-based Pro-forma model (forecasting of Revenue, COGS, SGA, Financing Costs, Taxes, WC, PPE, CF)
 - Behavioural Biases (Overconfidence, Illusion of control, Conservatism, Confirmation, Representativeness bias)
 - Porter's 5 Forces - B.P.o.Supplier (most impact on input costs), B.P.o.Customers, T.o.New Entrants, T.o.Substitutes, Industry Rivalry
 - Forecasting under -
 - Inflation (tackled using hedging via derivative contracts, vertical integration)
 - Industry pricing behaviours (Increasing prices - 1st mover vs laggard)
 - Margins under increasing costs
 - Price elasticity of demand effects
 - Forecast Horizon
 - Forecast terminal value
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Equity Investments - (11–14%, ~22 Qns)

39 - Market Organization & Structure

- Terminologies: Equilibrium Interest Rate
- Securities
 - **Equity** (common stock, preferred stock, warrants)
 - **Debt/FIS** (ST, LT, MT FIS; Bonds, Notes; Commercial Papers, Government Bills, Certificates of Deposit; Repurchase agreements, convertible debts)
- Pooled Investment vehicles (MFs, ETPs, HFAs, ABS)
- Currencies (5 Reserve currencies)
- Contracts (Forwards, Futures, Options - call & put, Swaps - currency, interest rate, equity, CDS)
- Terminologies: Money Market, Capital Market
- Brokers, Dealers, Exchanges
- Securitization
- Insurance risks (Moral Hazard, Adverse Selection, Fraud)
- Arbitrageurs
- Clearinghouses
- Custodians
- Long position
- Short position
- Hedging (e.g., wheat farmer)
- Short selling (covering short position, payment-in-lieu, short rebate rate)
- Long-short, call-put quadrant
- Margin call price (initial margin, maintenance margin, call money rate)
- Financial Leverage
- Sums - Margin Call Price, Margin Transactions
- Bid & Ask prices
- Market quotation (+ for liquid securities)
- Order execution types (market orders, limit orders)
- Market makers & price takers
- Limit order types & terminologies (Marketable/Aggressive, Inside the market/new market, make the market/at the market, behind the market, from the market)
- Order volumes (all-or-nothing, minimum size)
- Order visibility (hidden, iceberg)
- Order validity (day order, good-till-cancelled, immediate-or-cancel/kill-or-fill, good-on-close, good-on-open)
- Stop loss (buy, sell)
- Primary capital markets (IPOs, Seasoned offerings/Secondary issues), Secondary markets
- IPO terminologies (book building, accelerated book building, underwriting offering, best-effort basis, IB-issuer conflict)
- Private placements

- Shelf registration
- Dividend Reinvestment Plan
- Rights issue
- Market types (Call market, continuous market)
- Market structures
 - Quote-driven
 - Order-driven
 - Order matching - price priority (highest-bid, lowest-ask), time/display
 - Trade pricing rules - uniform, discriminatory, leverage
 - Brokered
- Market transparency
 - pre-trade (quotes & orders)
 - post-trade (price & volume)
 - Buy-side (transparency) vs dealers (opaque markets)
- Market efficiency (operationally, informationally, allocationally)

40 - Security Market Indexes

- Price-weighted index
- Equal-weighted index
- Market capitalization-weighted index
- Sums - price, equal & market capitalization
- Float-adjusted Market capitalization-weighted index
- Fundamental-weighted index
- Rebalancing
- Reconstituting
- Uses of security market index (market sentiment, benchmark, market R-R measure, input for beta & CAPM, model portfolio for index funds)
- Types of equity indexes (Broad-market, multi-market, Multi-market with fundamental-wi, sector, style)
- Fixed income indexes (cons - huge universe, high turnover, dealer-based trading, replication of FIS index)
- Alternative Investment indexes
 - Commodity
 - Real Estate
 - Hedge fund (Equally-weighted)

41 - Market Efficiency

- What is an informationally efficient capital market (i.e Efficient Markets Hypothesis)
- Investment strategy for Perfectly & Imperfectly efficient market
- Market value, intrinsic value

- Factors affecting Market Efficiency
 - Number of Participants
 - Availability of Information
 - Equal access to Information
 - Impediments to Trading (Transaction costs, Restrictions on Short Selling)
 - Arbitrage, Short Selling
- Fama's Forms of Market Efficiency
 - Weak
 - Semi-strong (Developed nations)
 - Strong
- Event study
- Market Anomalies
 - Time-series Anomalies
 - January (turn of the year) effect, Turn of the month, Day of the week, Weekend effect, Holiday effect
 - Overreaction
 - Momentum
 - Cross-sectional Anomalies
 - Size effect, Value effect
 - Other Anomalies
 - Closed-end funds, earnings announcements, IPO anomaly, and economic fundamentals
- Behavioral Biases
 - Selection, Overconfidence, Loss Aversion

42 - Overview of Equity Securities

- Common Shares characteristics -
- Voting system - statutory, cumulative
- Preference Shares characteristics
- Types of P.S.
 - Cumulative
 - Non-cumulative
 - Participating
 - Non-participating
 - Convertible
- Share Classes
- Private Equity
- Types of PE - VC, LBO & PIPE
- Direct Investing
- Depository Receipt
- Sponsored DR, unsponsored DR
- Global DRs

- American DRs
- Globally Registered Shares
- Basket of Listed DRs
- Risk-Return hierarchy
 - Cumulative - non-cumulative
 - Preferred stock - common stock
 - Puttable - callable
- Market value, book value of equity
- Price-to-Book ratio
- Return on equity

43 - Company Analysis: Past & Present

- Initial company research report
- Business model determinants (company information, public information, third-party information & primary research)
- Revenue drivers (top-down analysis, bottom-up analysis)
- Pricing Power & Market Competition (Low, High)
- Cost types (Operating, Financing & Investing)
- Operating Profit
- Contribution Margin
- Degrees of Operating Leverage
- Economies of Scale, Scope
- Degrees of Financial Leverage
- Sums - DOL, DFL
- Net WC/Sales

44 - Industry & Competitive Analysis

- Purpose of I&CA
- Risk from Portfolio-perspective
- Steps of I&CA
- Commercial Industry Classification (GICS, ICB, TRBC), Classification criteria
- Business cycle sensitive (defensive, cyclical)
- Growth rate sensitive (Growth stage, Mature stage)
- ROIC (measure of industry profitability)
- HHI Index
- Porter's 5 Forces
- PESTEL Analysis (E - cyclical, structural; T - sustaining, disruptive)
- Competitive strategy types
 - Cost Leadership
 - Differentiation

- Focused

45 - Company Analysis: Forecasting

- 4 Key forecast objects (drivers, without drivers, summary, ad-hoc items)
- Forecast approaches
 - Historical Results
 - Historical Base Rate Convergence
 - Management Guidance
 - Analyst Discretionary forecast
- Forecast horizon
- Revenue Forecasting
 - Top Down
 - Bottom Up
 - Recurring & non-recurring items
- Forecasting of -
 - COGS/Gross Margin
 - SGA Expenses
 - Working Capital
 - Capital Investments for Tangible & Intangible Assets
 - Maintenance, Capital Expenditures for Growth
 - Capital Structure

46 - Equity Valuation

- Intrinsic Value vs Market Value
 - Equity Valuation Models -
 - **Discounted CF Model** - DDM & FCFE
 - **Multiplier/Market Multiples** Model - Price & Enterprise
 - Asset-based Model
 - Dividends - Regular, Special, Stock dividends
 - Stock split, reverse stock split, Share repurchase
 - Dividend Payment Timeline
 - Dividend Discount Model
 - One-year holding
 - Multi-year holding
 - Preferred Stock Valuation
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Fixed Income - (11–14%, ~22 Qns)

47 - Fixed Income - Instruments & Features

- Major fixed income instrument - loans, bonds
- Bonds & their maturities - money market, capital market & perpetual
- FRNs
- Bond yield → bond prices, maturity
- Bond indenture
- Sources of repayment for various bonds - sovereign, local government, corporate secured, and corporate unsecured bonds
- Covenants - affirmative, negative

48 - Fixed Income - Cash Flow & Types

- **Principal Repayment Structures**
 - Bullet structure
 - Fully amortizing structure (*equal amortization of principal*)
 - Partially amortizing structure
 - Sinking fund provision (*least credit risk*)
 - Waterfall structure
- **Coupon Structure**
 - Fixed (*faces IR risk*)
 - FRNs (*decreases IR risk*)
 - Step-up (*protects against rising IR*)
 - Credit/Performance-linked
 - Payment-in-kind (PIK)
 - Green ESG bonds
 - Zero coupon
 - Deferred coupon
- **Index-linked bonds**
 - Interest-indexed (*principal unchanged, coupon varies*)
 - Capital-indexed (*principal varies, coupon fixed*) - *protects against IR variations*
- **Contingency Bond Provisions**
 - Callable - call protection, call schedule, *max price of callable = call value*
 - Puttable
 - Convertible
 - Warrants
 - Contingent Convertible (CoCo)
- **Bond types**
 - Domestic
 - Foreign
 - Euro

- Global
- International
- Sukuk
- Yield differences - currency denomination, local IR
- Tax treatment of bonds

49 - Fixed Income - Issuance & Trading

- Investment-grade, High-yield bonds
- F.I. product categorization - Maturity vs Credit quality
- Issuer categorization - Maturity vs Credit quality
- Primary market - underwriting offering, best-effort offering
- Shelf registration
- Secondary market
- Distressed debt
- Fixed Income index characteristics
- F.I. Benchmark selection criteria
- Index types - Aggregated (excludes junk bonds & unrated bonds), Specialized, ESG

50 - Fixed Income - Corporate Issuers

- External - Loan financing
 - Uncommitted LOC
 - Committed LOC
 - Revolving LOC
 - Factoring
- Commercial Paper
- ST Funding
 - Certificate of Deposits
 - Repurchase Agreement (Repo)
 - $\text{Purchase price (loan amount)} = \text{MV securities} / \text{initial margin}$
 - $\text{Repurchase price} = \text{Purchase price} \times (1 + \text{Repo} \times \text{adjusted for duration})$
 - $\text{Haircut} = 1 - 1/\text{I.M.}$
 - $\text{Variation Margin} = (\text{I.M.} \times \text{Repurchase price}) - \text{MV security}$
- Factors affecting Repo
- Repo Risks
- IG bonds vs HY bond
- Sums - Repo

51 - Fixed Income - Market for Government Issuers

61 - Credit Analysis - Government Issuers

- Sovereign debt issuers (*highest credit rating, largest debt issuers, public auctions, quote-driven OTC market, accounting method, economic balance sheet*)
- Sovereign market - Developed & Emerging - domestic debt & external debt
- Non-sovereign government bonds
 - Non-sovereign government bonds / Municipal bonds
 - Agency / Quasi-government bonds
 - General Obligation bonds
 - Revenue bonds
 - Supranational bonds
- Bid types - Non-competitive & competitive
- Auction types - single price & multi price
- On-the-run bond
- Sovereign bonds - creditworthiness assessment → 3 Quantitative & 5 Qualitative factors

52 - Fixed Income - Bond Valuation - Prices & Yields

- Valuing - Annual bond
- Valuing - Semi-annual bond
- Calculating - YTM (I/Y)
- Actual days method & 30/360 method
- Accrued Interest ($A.I = \text{Coupon Payment} \times (\text{days from last coupon to settlement} / \text{days in coupon period})$)
- Clean (flat) & Dirty (full) price
 - $\text{Full price} = \text{PV of bond on last coupon date} \times (1 + \text{YTM} / \text{No. of coupon payments in a year})^{\text{(days from last coupon to settlement} / \text{days in coupon period)}}$
- Sensitivity of Bond price → coupon rate to change in bond yield
- Sensitivity of Bond price → maturity to change in bond yield
- price-yield convexity, Constant-yield price trajectory
- Matrix pricing - similar bonds & spread

53 - Yield & Yield Spread Measures for Fixed-Rate Bonds

- Periodicity
- Annual Yield
- Current (income/running) Yield
- Simple Yield
- Yield to call, Yield to worst
- Callable bond (straight bond - call option)
- Yield spread (benchmark spread)
- G-spread
- I-spread

- Z-spread (*zero volatility spread, calculated based on shape of benchmark yield curve*)
- Option-adjusted spread (OAS)

54 - Yield & Yield Spread Measures - Floating-Rate Bonds

- Coupon rate formula ($FRN = MRR + QM$), quoted margin
- YTM formula ($FRN = MRR + DM$), discount/required margin
- FRN when $QM = DM$, $QM > DM$, $QM < DM$
- Valuation of FRN
- Add-on-yield for 365 days (Bond-equivalent yield - BEY)
- Holding Period Yield (HPY)
- Quoted Discount Yield

55 - Term Structure of Interest Rates - Spot, Par & Forward

- Valuing a bond using spot rates
- Par yield
- Forward Rates
 - Calculating Spot Rate from Forward Rate
 - Calculating Forward Rate from Spot Rate
 - Bond value using Forward Rate
- Spot Curve
- Yield Curve
- Par Yield Curve (*par yields slightly above spot rates*)
- Forward Curve

56 - Interest Rate - Risk & Return

- 3 sources of returns
- Reinvestment Risk vs Price Risk
- Macaulay duration
- Duration gap
- Sum - Capital G/L on a bond
- Sum - Investment Horizon Yields
- Sum - Interest Reinvested
- Sum - Macaulay duration using weighted averages

57 - Yield-based Bond Duration & Properties

- Modified Duration w.r.t Macaulay

- Approximate % change in a bond's price
- Approximate Modified Duration
- Money Duration
- Price Value of Basis Point
- Effect of Maturity, Coupon Rate, and Yield Curve on IR Risk
 - *Zero-coupon bonds* → *maximum duration*
 - *Discount bond* → *Macaulay duration increases initially, then decreases*
 - *Coupon bond* → *Duration decreases smoothly (if YTM constant)*

58 - Yield-based Bond Convexity & Portfolio Properties

- Convexity
- Convexity of a coupon-paying bond → Convexity for a CF at time t
- Approximate Convexity
- Convexity - Duration - Maturity - Coupon - YTM relationship
- % change in bond's price
- Duration effect, convexity effect
- Money convexity
- Change in bond's price
- Portfolio Duration

59 - Curve-based & Empirical Fixed Income Measures

- Bonds with embedded options vs Shift in benchmark yield curve
- Effective duration
- Effective convexity
- Convexity of option-free, callable, puttable bond
- Duration of option-free, callable, puttable bond at lower & higher yield
- Change in bond's price using effective duration & convexity
- Key rate duration

60 - Credit Risk

- What is credit risk
- 5 Cs of CR Analysis (Bottom-up approach) - Capacity, Capital, Collateral, Covenant, Character
- Top-down approach - Conditions, Country, Currency
- Collateral for secured corporate, unsecured corporate, and sovereign debt
- Expected Loss ($PD \times LGD$)
- LGD % ($Expected Exposure \times (1 - Recovery Rate)$)
- Credit spread ($PD \times LGD \%$)

- IG bond vs HY bond - unsecured/secured, yield spread, expected loss, spread volatility, spread dispersion
- IG & HY - Credit curves
- Credit Risk & Credit Spread as per Economic cycle (growth vs recession)
- Reason for increase in yield/credit spread
- Market liquidity risk
- Bid-offer spread
- Wide vs narrow spread (*narrow spread has low liquidity risk, higher credit quality, actively traded, better credit cycles & economic conditions, demand for debt*)
- Liquidity yield

62 - Credit Analysis for Corporate Issuers

- Qualitative factors affecting Credit Rating of Corporate Issuers
- Covenants - IG & HY bonds
- Quantitative factors - IG & HY bonds
- Credit Quality of Corporate Issuers
- Ratios used for Corporate Issuers (Profitability, Coverage, Leverage)
- Seniority Ranking of debt repayment
- Pari passu
- CFR, CCR
- Notching

63 - Fixed Income Securitization

- Securitization structure
- Seller, issuer, servicer, trustee (disinterested)
- Benefits of securitization to investors
- Risks to investors
- Benefits to markets
- Benefits to issuer

64 - Asset Backed Securities

- Covered bonds (*senior debt obligations, no SPE creation, dual recourse, LTV & overcoll*)
- Covered bonds vs ABS (*SPE, Tranching, static/dynamic pool*)
- Default structures - hard-bullet, soft-bullet, conditional pass-through
- Internal credit enhancements of ABS -
 - Overcollateralization
 - Excess spreads
 - Credit tranching

- Credit card ABS
- Collateralized Debt Obligation - CDO, CLO, CMO
- CDO Types - CF, Market value, synthetic

65 - Mortgage-Backed Securities

- Prepayment Risk (Contraction Risk)
 - Extension Risk
 - Interest Rate impact on Repayment
 - MBS convexity (*negative at lower yield*)
 - Time Tranching (*tackles prepayment risk*)
 - Residential MBS
 - Characteristics - prepayment penalties, recourse/non-recourse loans
 - Measures of default risk - LTV, DTI
 - Prime/sub-prime loans
 - Agency MBS & Non-agency MBS
 - Credit Rating of AMBS, NAMBS
 - Mortgage Pass-through securities
 - Define - CF - *admin fees*
 - PTR vs WAC
 - WAC, WAM
 - Collateralized Mortgage Obligation (CMO)
 - Sequential pay CMO
 - Z-tranches (Accrual)
 - Principal-only
 - Interest-only
 - Floating tranches
 - Residual tranches
 - PAC collar
 - Commercial MBS (CMBS)
 - CMBS vs RMBS - collateral pool size, diversification, loan repayer
 - WAMP
 - LTV, DSCR
 - Call protection
 - Loan-level → *prepayment lockout & penalty*
 - CMBS-level → *Sequential pay tranching*
 - Balloon Payment & Risk (*extension risk*)
-

Derivatives - (5–8%, ~12 Qns)

66 - Derivative Instruments & Market Features

- What is a derivative?
- Derivative terminologies - forward price, settlement/maturity date, contract/lot size
- Scenario - Spot price > Forward price
- Contract types - deliverable, cash-settled
- Hedging
- Why derivative market over cash market?
- CCH (Central Clearing House)
- Customized vs Standardized Contracts
- OTC vs Exchange Traded Markets

67 - Forward Commitment and Contingent Claim Features & Instruments

- Forward contract
- Futures contract
- Terminologies - Margin on Futures Exchange, MtM, Margin call in equity, Netting, CDS
- Swaps - IR
- Call Option - Buyer & seller/writer
- Put Option - Buyer & seller/writer
- Options Tree (Bull/bear, Pay/receive premium, Max profit, Max loss, Breakeven Point)
- Forward Commitment (Forwards, Futures, Swaps) vs Contingent Claim (Options, CDS)

68 - Derivative: Benefits, Risk, Issuer & Investor Uses

- Benefits
 - Risk Management (transfer of risk exposure without cash market transaction, create new risk exposures)
 - Information Discovery (market expectations)
 - Operational benefits (low transaction costs, higher liquidity, greater leverage, better price discovery, ease of short selling i.e. underlying not bought)
- Risk
 - Implicit Leverage
 - Basis risk (imperfect hedging w.r.t. underlying asset or timing differences of exposure)
 - Liquidity risk
 - Counterparty risk
 - Systemic risk
- Issuer Uses

- Corporate Users - exposure to FX, IR, commodities; hedging FX income, inventory value
- Hedge Accounting - Cashflow hedging (domestic currency hedged using a forward contract, Fair value (offsets changes in Asset/Liab. fair values), Net investment (value of foreign subsidiaries on BS hedged))
- Investor Uses
 - Hedging (limiting downside)
 - Speculation/Greater Leverage
 - Risk Modification

69 - Arbitrage, Replication & Cost of Carry

- Arbitrage & No-Arbitrage
- Forward price - No-Arbitrage relationship on Forward contracts & shares/underlying
- Formula - No-Arbitrage Forward Price
- Replication
- Formula - No-Arbitrage Forward Price with Costs & Benefits
- Net Cost of Carry
- Formula - Continuous compounding - No-Arbitrage Forward Price (Basic, Costs & Benefits)
- Sum - Continuous compounding
- Forward Contracts on Currencies

70 - Pricing & Valuation of Forward Contracts

- Price, Value of a forward contract
- Contango & Backwardation
- Forward Rate Agreement (notation, sums)
- Implied Forward Rate
- Formula - Value of Forward contract - Initiation, during its life, expiration (buyer & seller)
- Formula - Value of Forward contract (incl. costs & benefits)

71 - Pricing & Valuation of Futures Contracts

- Interest Rate Futures Price
- Basis Point Value
- Correlation between Interest Rates & Futures Prices and Impact on Futures & Forwards Prices
- Convexity & Convexity Bias

72 - Pricing & Valuation of IR & Swaps

- Structure of a Swap
- Swap - FRAs relationship
- Par Swap Rate
 - (Borrow at Fixed Rate, Lend at MRR)
- Price of Swap
- Value of Swap
- Impact of IR changes on Fixed-Rate Swap payer

73 - Pricing & Valuation of Options

- Moneyness
- Call option → In-the-money, At-the-money, Out-of-the-money
- Put option → In-the-money, At-the-money, Out-of-the-money
- Option Premium
 - Exercise value/Intrinsic value
 - Time value/Speculative value
- European Options
- Lower (Min) & Upper (Max) bounds of European Options - Call & Put
- Impact of Variables on Values of Call & Put Options
 - Volatility
 - Time to expiry
 - Spot price
 - Exercise price
 - Risk-free Rate
 - Benefits (dividends, interest earned)
 - Costs (holding costs, storage costs)

74 - Option Replication Using Put-Call Parity

- Put-call parity
- Fiduciary Call
- Protective Put
- Put-call parity payoff when $S \geq X$ and $X > S$
- Synthetic payoff on Long Stock
- Sum - Put-call parity
- Put-call-forward parity
- Put-call parity w.r.t. Firm's Assets

75 - Valuing Derivatives using One-Period Binomial Model

- Binomial Model
 - Constructing a one-period binomial model
 - Return on up-move & down-move (R_u , R_d)
 - Binomial Tree
 - Call value up-move & down-move (C_u , C_d)
 - Value of call option ($V_u = V_d$)
 - Discounted value of call option (V_0)
 - Initial value of call option (C_0)
 - One-period binomial model with risk-neutral probabilities
 - Risk-neutral probability formula
 - Expected call value in 1 year using R-N formula
-

Alternative Investment - (7–10%, ~15 Qns)

76 - Alternative Investment - Features, Methods & Structures

- **About Alternative Investments** (*actively managed, risk reduction via diversification, low correlation with traditional investments, higher returns | direct investments, information asymmetry, difficult to evaluate performance, less historic data, data volatility*)
- **Alternative Investments vs Traditional Investments** (liquidity, time horizon, size of commitments, correlation with TI, specialized knowledge, management fees)
- **3 categories of Alternative Investments**
 - Private Capital - Private Equity & Private Debt
 - Real Assets - Commodities, Timberland, Farmland, Intangible (Patents, wills), Art, Digital Assets (Crypto)
 - Hedge Funds
- **3 types of investing in AI**
 - Fund Investing
 - Co Investing
 - Direct Investing
- Alternative Investment Structure (*LP-GP*)
- Terminologies - Side letters & Most-Favourable-Nation clause, Master LPs, Committed Capital, Catch-up clause, High watermark, Clawback provision
- Fee Structure
 - Management Fees
 - Performance Fees
- Hurdle Rates (*hard, soft*)
- Waterfall Structure
 - American (Deal by deal)
 - European (Whole of funds)

77 - Alternative Investment - Performance & Returns

- **Alternative Investment Risks** (*CF Timings, Use of Leverage, absence of market valuations, complexity - fees, taxation, accounting*)
- **Risk 1 - Three phases of CFs Timings -**
 - Capital Commitment
 - Capital Deployment
 - Capital Distribution
- J-curve
- Measure of Alt. Investment CF Performance - IRR, MoIC (appropriate for, drawback)
- **Risk 2 - Use of Leverage**
 - Unleveraged, Leveraged returns formula
 - Risk of using Leverage
- **Risk 3 - Valuation of Alt. Inv.**

- 3-level valuation hierarchy - market valuation, valuing via observable inputs, valuing at cost
- Fee Structure Terminologies -
 - Lockup period
 - Notice period
 - Gate
 - Founder class shares
 - Annual investor fees
- Biases in AI - survivorship, backfill bias
- Sums - Fund-of-Funds Fees, Hedge Fund Fees, Waterfall Structure

78 - Investment in Private Capital - Equity & Debt

- Portfolio companies
- Leveraged Buyouts (+ MBOs, MBIs)
- Venture Capital
- **VC Investment Stages in a company** (Pre-seed/Angel, Seed capital, Early/Startup, Later stage/VC Expansion, Mezzanine)SPACs,
- Private Investment in Public Equity (PIPE)
- PE Exit Strategies (Trade Sale, IPO, Direct Listing, SPAC, Secondary Sale, Recapitalisation, Liquidation)
- Private Debt types (Direct lending, VC debt, Distressed debt, Mezzanine debt, Unitranche debt)
- Cos. to invest in as per the Business cycle
- Risk-Return Profile of Private Capital (*private equity > mezzanine debt > unitranche debt > senior direct debt > real estate debt > infra debt*)

79 - Real Estate & Infrastructure

- Real Estate Quadrant
 - Private - Equity (*privately owning buildings via a RE PE fund*)
 - Private - Debt (*lending privately against RE - lending loan to property developer*)
 - Public - Equity (*owning RE via stock market*)
 - Public - Debt (*investing in bonds backed by RE*)
- Direct Ownership of R.E.
- Indirect Ownership of R.E.
- REITs (*no double taxation, no redemption/liquidation risk*)
- R.E. strategies (core, core-plus, value-added, opportunistic)
- Risk Profile of RE strategies
- Types of Infrastructure Investments (Transportation, Utility, Social infra., Info-Comm. Tech.)

- Cash Flows from Infra Assets (Available payment, Usage-based payment, Take-or-pay arrangement)
- Greenfield Infrastructure Investment
- Brownfield Infrastructure Investment
- Secondary-stage Investment
- Cons of investing in infrastructure

80 - Natural Resources

- Raw land, Farm land, Timber land (liquidity, income, investors)
- Commodity sectors (metals, agri products, energy products)
- Commodity contracts (grade, delivery location)
- Exposure to Commodities
 - Exchange traded
 - Managed future funds (*actively managed, sector focus/diversified, structured as LPs or MFs → SMAs, Commodity Trading Advisors, Specialized funds*)
 - Specialized funds
- Commodity Valuation (*future price = spot price + net cost of carry; future price = spot price $\times$ (1+Rf) + storage cost - convenience yield*)
- Contango, Backwardation
- Commodity Correlation with Stocks, Bonds & Inflation

81 - Hedge Funds

- HF characteristics (*private-pooled investments, qualified investor only, less regulated, hybrid investment strategies, high cost, high fees, low liquidity, low transparency, ST horizon, periodic redemptions, primary returns, performance evaluation criteria*)
- HF strategy categories -
 - Equity
 - Fundamental Long/Short
 - Fundamental Growth
 - Fundamental Value
 - Market Neutral
 - Short Bias
 - Event-driven
 - Merger arbitrage
 - Restructuring/distressed
 - Activist SH
 - Special Situations
 - Relative-value
 - Convertible arbitrage fixed income
 - General fixed income

- Special fixed income
 - Multistrategy
 - Opportunistic
 - Macro
 - Managed Future Funds
- HF structures
 - Commingled (+ Master-feeder)
 - SMA
 - LP-LLP
- Fund-of-Funds
- HF Returns
 - Market Beta
 - Strategy Beta
 - Alpha
- HF Biases (survivorship, backfill, selection)

82 - Introduction to Digital Assets

- What are Digital Assets
- Distributed Ledger Technology (DLT)
- DLT Network (digital ledger, consensus mechanism, network participants)
- Smart contract
- Blockchain
- Consensus Protocol
 - Proof of Work
 - Proof of Stake
- DLT Network Types
 - Permissionless
 - Permissioned
- Types of Digital Assets
 - Cryptocurrency
 - Alt coins
 - Stable coins
 - Meme coins
 - CBDC
 - **Tokenization**
 - NFT (*certificate of authenticity*)
 - Security token (*publicly-traded securities*)
 - Utility token (*network services*)
 - Governance token (*voting*)
- Digital asset vs Traditional asset (inherent value, transaction validation, regulations, medium of exchange)
- Types of Crypto Exchanges

- Centralized
 - Decentralized
 - Investment in crypto
 - Direct
 - Indirect
 - Coin Trust
 - Futures contract
 - ETPs
 - Stocks
 - HFs
 - Asset-backed tokens
 - DeFi
-

Portfolio Management - (8–12%, ~18 Qns)

83 - Portfolio Risk & Return - Part 1

- 3 Investor Types as per Risk-taking (Risk Averse, Risk Seeking, Risk Neutral)
 - Risk Averse - *Risk-Return characteristic is positive; More the RA, Lesser the utility & expected returns*
- Investor Utility Function (*it is 0 for a risk-free asset*)
- Indifferent Curve (*most RA, most convex curve*)
- Indifferent Curve for Risk Averse Investor
- Capital Allocation Line ($R_f + \text{Risky Assets}$ - *Optimal Portfolio*)
- Sums - Expected Return of Portfolio
- Sums - Standard Deviation of Portfolio
- Two-Fund Separation Theorem
- Sums - Variance (Sample, Population)
- Sums - Covariance (Sample)
- Sums - Covariance (*Contributes the most to volatility in a portfolio*)
- Sums - Correlation
- Efficient Frontier

84 - Portfolio Risk & Return - Part 2

- Capital Allocation Line
- Capital Market Line (& Equation)
- Lending & Borrowing Portfolios
- Total Risk
- Systematic & Unsystematic Risk
- Return generating models
- Single-index model
- Market model
- Beta
- Sums - Calculating Beta of an Asset
- Security Market Line
- CAPM
- Sums - Mispricing of securities
- Portfolio Performance basis Total Risk
 - Sharpe Ratio
 - M-squared
- Portfolio Performance basis Systematic Risk
 - Treynor Ratio
 - Jensen's alpha

85 - Portfolio Management: An Overview

- Portfolio perspective
- Diversification of portfolio
- Modern Portfolio Theory
- Diversification Ratio
- Portfolio Management process (Planning, Execution, Feedback)
- Investment Policy Statement
- Top-down & Bottom-up Analysis
- Types of Investment Management Clients - (Individuals, Banks, MFs, Insurance, Endowments & Foundations)
- Defined Pension plans - Contributions, Benefits
- Buy & sell side
- Active & Passive Management Investing
- Types of Asset Managers (Full-service, Specialist; Traditional & Alternative)
- Mutual Funds - Terminologies (NAV, Load, No-load)
- Open & Closed-end funds
- Types of MFs
- ETFs
- Separately Managed Accounts
- Hedge Funds

86 - Basics of Portfolio Planning & Construction

- Investment Policy Statement definition
- Investment Policy Statement Components
- Risk Objectives (Absolute & Relative)
- Financial Risk Tolerance (Ability & Willingness)
- Investment Constraints (RRTLLU)
- Strategic Asset Allocation
- Tactical Asset Allocation
- Security Selection
- Core-Satellite approach
- ESG considerations - Negative screening, Positive screening, Thematic, Impact, Engagement/Active ownership, ESG integration

87 - Behavioural Biases of Individuals

- Cognitive Errors
 - Belief Perseverance
 - Conservatism
 - Confirmation

- Representativeness (base-rate neglect, sample-size neglect)
 - Control (illusion of knowledge, self-attribution, overconfidence)
 - Hindsight
 - Information Processing
 - Anchoring & adjustment
 - Mental Accounting
 - Framing
 - Availability
- Emotional Biases
 - Loss Aversion
 - Overconfidence (self-enhancing, self-protecting, prediction overconfidence, certainty overconfidence)
 - Self control
 - Status Quo
 - Endowment
 - Regret-aversion (error of commission & omission)
- Halo effect
- Home bias

88 - Introduction to Risk Management

- What is Risk Management
- Risk Management Framework
- Risk Governance
- Risk Tolerance
- Risk Budgeting
- Financial Risk (Credit, Liquidity, Market)
- Non-Financial Risks (Operational, Regulatory, Political, Legal, Solvency, Model, Tail, Accounting)
- Measures of Risk (Std. dev., Beta, Duration)
- Derivative Risks (Theta, Rho, Vega, Delta, Gamma)
- Value at Risk
- Conditional Value at Risk
- Stress Testing
- Scenario Analysis
- Self-insurance
- Risk Transfer (insurance), surety bonds, fidelity bonds
- Risk shifting (derivatives)

Ethics - (15–20%, ~30 Qns)

89 - Ethics & Trust in Investment Profession

- Suitability standard
- Fiduciary standard
- Ethics vs code of ethics vs standards of conduct
- Challenges to ethical behaviour (*overconfidence, ST focus, situational & external influences*)
- Impact of situational & external factors on ethical conduct
- Profession
- MCQs

90 - Code of Ethics & Standards of Professional Conduct

- 7 Standards of Professional Conduct -
 - Professionalism
 - Duty to clients
 - Duty to the employer
 - Integrity of capital markets
 - Conflict of interest
 - Investment Analysis, Recommendations & Actions
 - Responsibilities as a CFA member

91 - Guidance for Standards I-VII

- **Professionalism**
 1. **Knowledge of the Law** (*obey, report violations, dissociate*)
 2. **Independence & Objectivity** (*gift, benefits & compensation acceptance and its impact on I&O, conflict management, paid travel policies, personal investment policies*)
 3. **Misrepresentation** (*false statements, misleading claims, material omissions, plagiarism*)
 4. **Misconduct** (*fraud, deceit, cheating, dishonesty, failure to perform daily BAUs diligently*)
 5. **Competence** (*skill, knowledge & diligence*)
- **Integrity of Capital Markets**
 - **MNPI** (*equitable information dissemination, insider trading via MNPI dissemination, mosaic theory*)
 - **Market Manipulation** (*artificial price or volume distortion*)
- **Duties to Employers**

- **Loyalty** (*perform to the best of skills & abilities, protect confidential information, do not cause harm, do not engage in competitive activities, take permission for independent practice, whistleblowing encouraged*)
- **Additional Compensation Arrangements** (*can receive direct or indirect gifts if written consent*)
- **Responsibilities of Supervisors** (*set expectations, detect & prevent misconduct, ensure robust compliance*)
- **Duties to Clients**
 - **Loyalty, Prudence & Care** (*act in client's interest, take decisions as if it's your own asset, align client-investment objectives, soft commission policies, proxy voting*)
 - **Fair Dealing** (*fair != equal, fairness in recommendations, different services to different clients, equitable information dissemination, IPO-secondary offerings*)
 - **Suitability** (*IA,R,A as per client's IPS, trade request's impact on risk-return profile*)
 - **Performance Presentation** (*misrepresentation prohibited, past performance presentation, use GIPS, tailor PPT as per audience, include past accounts*)
 - **Confidentiality Preservation** (*preserve unless illegal, by law, or allowed by client*)
- **Information Analysis, Recommendation & Actions**
 - **Due Diligence & Reasonable Basis** (*provide I-A,R only after proper research, consider EIC factors, required to scrutinize third-party reports & assumptions, recommended to substantiate, quant model testing*)
 - **Communication with Clients & Prospective Clients** (*disclose what you do & don't, nature of product, cost associated with it, investment process, risks, limitations, what services will be provided & will not be, reasonable judgement*)
 - **Record Retention** (*for at least 7 years*)
- **Conflict of Interest**
 - **Avoid or Disclose conflicts** (*conflict reasons like compensation structures, personal investments, external relationships, competing duties to clients; disclose everything to clients like personal/beneficial ownership in stocks, direct/indirect control of vote in securities, position on boards, market making activities, etc.*)
 - **Priority of Transactions** (*always client first, personal investing is not unethical as long as client needs are kept first, beneficial ownership, limited participation in markets - IPO, private placements, blackouts, pre-clearances*)
 - **Referral Fees** (*disclose referral fees, clarify the nature of compensation*)
- **Responsibilities as a CFA Member**
 - As a CFA candidate/member, as a CFA holder (*exam-related content like questions, topics tested should be confidential, discussion of curriculum & non-confidential information and expression of personal opinion is allowed, CFAI membership criteria*)
 - When referencing the CFA Institute, membership, or candidacy, members must not misrepresent or exaggerate the meaning or implications of the designation (e.g., promising superior performance).

92 - GIPS

- What is GIPS
- GIPS 8 standards for firms
- Composite
- Discretion
- Independent Verification

ALL THE BEST!